Student Guide — primary / module-11-multiplication-near-base

IKS Curriculum

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Student Workbook

Student Workbook — Multiplication Near a Power of 10 — Nikhilam (Primary)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Nikhilaṁ* — ___________________________
- *vinkulam* — ___________________________
- *ādhikya* — ___________________________
- *Ūrdhva-tiryagbhyāṁ* — ___________________________

Day-by-day notes

Day 1 — Finger-Trick Prelude

What I learned:

One question I have:

Day 2 — Two-Digit × Two-Digit Near 100 (Below)

What I learned:

One question I have:

Day 3 — Two-Digit × Two-Digit Near 100 (Above)

What I learned:

One question I have:

Day 4 — Mixed — One Above, One Below

What I learned:

One question I have:

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One question I have:

Day 9 — Mixed Practice + Module 7 Crossover

What I learned:

One question I have:

Day 10 — Final Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — Make the Twin Game

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children play a card game finding the "twin" — the partner that makes 10 with each card.

Materials

- A deck of number cards 1–9 (4 of each = 36 cards) per pair.
- Counters (optional).

The flow

Settle + introduce the chant (5 min)

Teach the chant — clap on each pair:

> "1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!"

Say it 3 times. (Children from Module 5 will know this — re-use it.)

Story (5 min)

"Today we play a game called Make the Twin. Every number from 1 to 9 has a twin that makes 10 with it. 1's twin is 9. 2's twin is 8. Today we find them."

Game (17 min)

How to play:

1. Shuffle the deck. Deal 5 cards to each player. Place 5 cards face-up in the middle.
2. On your turn:
   - Look at your hand and the middle.
   - Find a "twin pair" — your card + a middle card that add to 10.
   - Take both. Put them in your "pile."
   - Draw a new card from the deck.
3. If you can't find a twin, place 1 card in the middle. End your turn.
4. When the deck is empty, count pairs. Most pairs wins!

Play 2 rounds.

Show + chant (3 min)

- "Who won a round?" — winner shares ONE twin pair they made.

- Class chants the pair: *“6 and 4 make 10!”*
- Close with the daily chant.

Homework

- At home, practise the chant with a family member. Win a round of Make the Twin if you can find a partner!

Differentiation

- **Tier up:** Add cards 0 and 10. Find twins for those too. (0 + 10. Or just play with 1–9.)
- **Tier down:** Use only cards 1–6 to keep the twins easy.

Teacher’s quick notes

- Some children won’t find twins and will give up. Pair them with a stronger player as a “team.”
- The game IS the lesson at this band. Don’t add a worksheet.
- Note: children who completed Module 5 will be fast at twins. Push them with the “speed challenge” — name the twin in under 1 second.

Day 2 — Twins Chant

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children commit the 9 deficit-from-10 pairs to memory and can recall any twin in under 1 second.

Materials

- Class wall-poster: “Twins of 10” (pre-made; show all 9 pairs).
- Index cards with single digits 1–9 (one set per child).
- Stopwatch.

The flow

Settle + chant (5 min)

Sing the chant:

> “1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!”

Three times, with claps.

Now add the “deficit” version (just renaming):

> “1’s twin is 9! 2’s twin is 8! 3’s twin is 7! 4’s twin is 6! 5’s twin is itself!”

Story (5 min)

- *“Today we make the twins so fast we don’t even think. We just KNOW.”*
- Hold up the wall poster. Walk through the 5 pairs. Notice: each pair appears twice in the “twin” naming — 1’s twin is 9 AND 9’s twin is 1.

Lightning round (10 min)

- Teacher holds up a digit. Class chorus the twin.
- Do 20 in 60 seconds.
- Repeat with random sequence.

Sequence example:

```
7 → 3
8 → 2
6 → 4
9 → 1
5 → 5
4 → 6
3 → 7
2 → 8
1 → 9
8 → 2
6 → 4
7 → 3
...
```

Card sort (7 min)

- Each child gets index cards 1–9.
- On “Go,” they arrange the cards into 5 pairs (with 5’s pair being itself or stand-alone).
- Fastest 3 children share their arrangement on the floor.

Show + chant (3 min)

- Final chant. Pull the index cards back into the deck.

Homework

- At home, have someone call out 10 digits in 30 seconds. You say the twin. Beat 8/10.

Differentiation

- **Tier up:** Speed of 30 twins in 30 seconds.
- **Tier down:** Slow chant. 5 twins in 30 seconds is fine for today.

Teacher’s quick notes

- Pay attention to children who PAUSE before answering. They haven’t built the reflex yet. Pull them into more drills tomorrow.

- The single most important fluency in this module. Don't move on if half the class is still pausing.

Day 3 — The Finger Trick

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children learn the hand-multiplication trick for any pair from 6 to 9.

Materials

- Both hands (everyone has them).
- Class wall-poster: a hand diagram showing fingers folded down for `7 × 8`.
- Worksheet: 16 problems (all combinations 6×6 through 9×9), with hand-diagram answer key.

The flow

Settle + chant (5 min)

The Twins chant + lightning round of 10 twins.

Story (8 min) — "The Magic Hand Trick"

- *"Today I'll teach you a magic trick. You can multiply 7 × 8 just by counting fingers. Watch."*
- **Demo: 7 × 8.**
    - "Hold up both hands. We're going to fold down some fingers."
    - "For 7: fold down (7 − 5) = 2 fingers on the LEFT hand. So 2 down, 3 up on the left."
    - "For 8: fold down (8 − 5) = 3 fingers on the RIGHT hand. So 3 down, 2 up on the right."
    - "Count the fingers DOWN on both hands together: 2 + 3 = **5**. That's the TENS. → **50**."
    - "Count the fingers UP on left × fingers UP on right: 3 × 2 = **6**. That's the UNITS. → **6**."
    - "Total: 50 + 6 = **56**. ✓"
- Repeat for `6 × 9`:
    - 6: 1 down, 4 up. 9: 4 down, 1 up.
    - Down sum: 1 + 4 = 5 → 50.
    - Up product: 4 × 1 = 4.
    - Answer: **54**. ✓
- Repeat for `8 × 9`:
    - 8: 3 down, 2 up. 9: 4 down, 1 up.

    - Down sum: 3 + 4 = 7 → 70.
    - Up product: 2 × 1 = 2.
    - Answer: **72**. ✓

Practice (12 min)

- Whole class: teacher calls `7 × 6`. Children put up hands and find the answer (42).
- 5 problems chorus: `9 × 9` (81), `6 × 7` (42), `8 × 8` (64), `9 × 6` (54), `7 × 9` (63).
- Pair drill (last 7 min):
    - Partner A calls a problem (any from 6×6 to 9×9).
    - Partner B answers using the finger trick.
    - Swap.

Show + chant (5 min)

- "Who got 9 × 9 = 81 using their hands?" — hands up.
- Final chant.

Homework

- Teach the finger trick to a parent or sibling. Multiply `7 × 8` together.

Differentiation

- **Tier up:** Try `5 × 6` — what does the trick give? (5: 0 down, 5 up. 6: 1 down, 4 up. Down sum 1 → 10. Up product 5×4 = 20. Total 30. ✓) Notice: works for 5+ too. What about `5 × 5`? (Both 0 down. Down sum 0. Up product 5×5 = 25. ✓ Total 25.) Cool — works for 5 too.
- **Tier down:** Stay with `7 × 8` and `9 × 9` until comfortable. Don't push variety today.

Teacher's quick notes

- A few children will have trouble keeping track of "down vs up." Slow down with them; the trick *is* tricky on the first day.
- Some children will object: "I already know my 7 times table." Honest response: *"Great! Now you have TWO ways of finding the answer."*
- Don't explain why it works today. Day 4 is the dot-drawing day.

Sanskrit term mention

(Hold for Day 6.)

Day 4 — Why the Trick Works (Dot Drawing)

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children draw a dot pattern showing why `7 × 8 = 56` and *see* the connection between fingers and the answer.

Materials

- A4 paper per child (or larger).
- Coloured crayons / pencils (2 colours per child works fine).
- The hand poster from Day 3.

The flow

Settle + chant (3 min)

Quick Twins chant + 5 finger-trick chorus answers.

Story (7 min) — “Let’s see WHY”

- *“Yesterday I told you the trick. Today I’ll show you why it works.”*

- Draw a `7 × 8` rectangle on the board: 7 columns, 8 rows.

- Count the total dots together: 7 columns × 8 rows = 56 dots.

- Now divide the rectangle into regions:

    - A **5 × 5 block** in one corner (the “fully solid” block).
    - A **5 × 3 strip** below (because 8 = 5 + 3).
    - A **2 × 5 strip** to the side (because 7 = 5 + 2).
    - A **2 × 3 corner** (fingers UP × fingers UP).

  Total dots: 25 + 15 + 10 + 6 = 56. ✓

- *“See the 2 × 3 corner? That’s the fingers UP — and 2 × 3 = 6, the units of our answer.”*

- *“See the 5 × 5 block plus the two strips? Together they make 50 — the tens of our answer.”*

Drawing (15 min)

- Each child draws a `7 × 8` dot rectangle on their A4 paper.
- Colour the 2 × 3 corner one colour (the “fingers up” region).
- Colour the rest a different colour (the “tens” region).
- Count and label: “fingers-up corner = ***” and ”rest =*** .”

Show + chant (5 min)

- 2 children share their drawings.
- Final chant.

Homework

- Pick another problem (6 × 7 OR 9 × 8) and draw the dot rectangle. Colour the corners. Bring it tomorrow.

Differentiation

- **Tier up:** Draw the same picture for `9 × 9`. The corner is 1 × 1 = 1 dot. The rest is 80 dots. Total = 81. ✓
- **Tier down:** Pre-printed grid; child just colours the regions.

Teacher's quick notes

- The drawing IS the proof. Children who see the picture once will trust the finger trick forever.
- Some children will try to colour outside the lines, or count dots wrong. That's fine — the colouring is the point, the counting is the practice.
- Don't introduce algebra. The dot drawing is the entire "why" for this band.

Day 5 — Pairs Race

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children speed up the finger trick to under 5 seconds per problem.

Materials

- A "Race Sheet" — 16 problems (all combinations 6×6 through 9×9) on a single page.
- Stopwatch.

The flow

Settle + chant (3 min)

Twins chant + 5 finger-trick chorus answers.

Race round 1 (8 min)

- Pass out the Race Sheet (face-down).
- On "Go," flip and solve all 16 problems using the finger trick. As fast as possible.
- After 3 minutes, "Stop." Note how far each child got and how many were correct.
- Self-correct against the answer key.

Race Sheet:

```
6×6  = ___        7×6  = ___        8×6  = ___        9×6  = ___
6×7  = ___        7×7  = ___        8×7  = ___        9×7  = ___
```

```
6×8 = ____     7×8 = ____     8×8 = ____     9×8 = ____
6×9 = ____     7×9 = ____     8×9 = ____     9×9 = ____
```

Answer key:

```
36  42  48  54
42  49  56  63
48  56  64  72
54  63  72  81
```

Race round 2 (7 min)

- Same sheet, fresh strip. Solve as many as possible in 2 minutes.
- Beat your round-1 count.

Pair quick-fire (8 min)

- Pair up.
- Partner A reads a problem from the sheet. Partner B answers within 5 seconds (using fingers).
- Swap. Cover 10 problems each.

Wrap-up (4 min)

- Hands up: who got all 16 right? Who beat their round-1 score?
- Cheering. Final chant.

Homework

- Practice 6 × 6 through 9 × 9 at home. 5 minutes a day. Time yourself.

Differentiation

- **Tier up:** Try problems with at least one factor being 5 (e.g. `5 × 8`, `5 × 9`). The trick still works! And: do all 16 in 60 seconds.
- **Tier down:** 8 problems instead of 16. Half the sheet is fine.

Teacher's quick notes

- Don't make this competitive between children. *"Race yourself, not your neighbour."*
- A child who got 4/16 should celebrate — that's progress from yesterday's 0/16.
- Some children will revert to memorised times-table answers. That's fine, but ask them to confirm with the finger trick.

Day 6 — Why It Works — The Algebra

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Derive $(B-a)(B-b) = B(B-a-b) + ab$.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *vinkulam* (IAST: vinkulam; lit. "deficit (number below base)")

- Why does the trick work? Let's see with small numbers!

Show 9 × 8 using the rule: - 9 is 10−1 (deficit 1). - 8 is 10−2 (deficit 2). - Left: 9 − 2 = 7. Right: 1 × 2 = 2. Combined: **72**. - Check: 9 × 8 = 72. ✓

"Same answer as tables! The trick is just another way."

Activity (8 min)

A simplified version of *Method Speed Race* (see `activities/`). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See `homework.md`.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — When To Use It

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Identify problems where Nikhilam is faster than long multiplication.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *ādhikya* (IAST: ādhikya; lit. "excess (number above base)")

- The trick is special! It works best when numbers are CLOSE to 100 (or 10, or 1000).

- For numbers far from 100 (like 47 × 63), we use the regular way.

Show 97 × 96 (use trick) and 47 × 63 (use regular). *"Pick the right tool!"*

Children practice sorting: which problems use the trick?

Activity (8 min)

A simplified version of *Method Speed Race* (see `activities/`). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See `homework.md`.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Speed Sprints

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Race traditional vs Nikhilam on 30 problems.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *Ūrdhva-tiryagbhyāṁ* (IAST: Ūrdhva-tiryagbhyāṁ; lit. ""vertically and crosswise" — the general-purpose sūtra")

- Speed game! 5 problems with the trick, 3 minutes.

97 × 96 = ? 98 × 97 = ? 99 × 99 = ? 95 × 95 = ? 103 × 104 = ?

(Teacher does these together with kids — they don't need to do them solo yet.)

Cheer at the end. *"Maths magic!"*

Activity (8 min)

A simplified version of *Method Speed Race* (see `activities/`). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See `homework.md`.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Mixed Practice + Module 7 Crossover

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Solve problems combining ×11 and near-base methods.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *Nikhilaṁ* (IAST: Nikhilaṁ; lit. "the Nikhilam sūtra (also used in Module 9 for subtraction)")

- Today: mix ×11 trick (Module 7) and the new trick!

Show 11 × 97 — use the ×11 trick: 1067. Show 97 × 96 — use the new trick: 9312.

Children sort: *"Which trick for which problem?"*

End with a number-magic round of applause.

Activity (8 min)

A simplified version of *Method Speed Race* (see `activities/`). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See `homework.md`.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.

- If energy is low, swap the activity for free drawing.

Day 10 — Final Assessment

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Summative quiz and method-comparison reflection.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *vinkulam* (IAST: vinkulam; lit. "deficit (number below base)")

- Final day! We celebrate the new trick.

Each child shows their best near-100 multiplication. Cheer.

Quiz with stickers — everyone gets a "Number Wizard" sticker.

End chant: *"Cross-subtract, multiply, put together!"*

Activity (8 min)

A simplified version of *Method Speed Race* (see `activities/`). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See `homework.md`.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.

- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Homework

Homework — Multiplication Near a Power of 10 — Nikhilam (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — Finger-Trick Prelude** · Draw about today's concept. (~5 min)
2. **Day 2 — Two-Digit × Two-Digit Near 100 (Below)** · Draw about today's concept. (~5 min)
3. **Day 3 — Two-Digit × Two-Digit Near 100 (Above)** · Draw about today's concept. (~5 min)
4. **Day 4 — Mixed — One Above, One Below** · Draw about today's concept. (~5 min)
5. **Day 5 — 3-Digit × 3-Digit Near 1000** · Draw about today's concept. (~5 min)
6. **Day 6 — Why It Works — The Algebra** · Draw about today's concept. (~5 min)
7. **Day 7 — When To Use It** · Draw about today's concept. (~5 min)
8. **Day 8 — Speed Sprints** · Draw about today's concept. (~5 min)
9. **Day 9 — Mixed Practice + Module 7 Crossover** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.